

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 8

1. **Hypothesis Testing:** to make a claim about a population parameter(s) or distribution(s), based on collected data.

- Hypotheses are claims about the population parameters.
- This time: one sample hypothesis test on the population mean μ .

2. **Null hypothesis and Alternative Hypothesis**

- Null hypothesis H_0 : the conservative claim
- Alternative hypothesis H_a : what the scientist wants to prove, a "radical" claim, etc
- Three cases: (i) $H_0 : \mu = \mu_0$; $H_a : \mu \neq \mu_0$; (ii) $H_0 : \mu = \mu_0$; $H_a : \mu > \mu_0$; (iii) $H_0 : \mu = \mu_0$; $H_a : \mu < \mu_0$.
- Some examples (For each of the following applications, state what a hypothesis test of interest would be, (null and alternative hypothesis))
 - (a) A new COVID-19 vaccine by Gilead, a pharmaceutical company, for the adult population in the US
 - (b) a machine learning algorithm that forecasts financial crisis (defined as a drop of the stock market index like S&P500 by more than 30%) using macroeconomic variables
 - (c) A well-known food brand (think Kraft or Cadbury) claims that there're 500g of chocolate pieces in the bag

– **Solution:**

- * A parameter of interest would be the efficacy rate, that is, the percentage of population for which the vaccine is effective against COVID-19. A possible alternative hypothesis would be that Gilead's vaccine's efficacy rate is higher than the vaccine out on the market earlier (e.g., Pfizer's). Scientific applications tend to be conservative and want to minimize type I error (e.g., that the innovation is not actually innovative and does not improve on the status quo)
- * A parameter of interest would be the proportion of financial crisis that is correctly forecasted by the algorithm. A alternative hypothesis would be if the algorithm can forecast more crisis correctly than a traditional method backed by economic theory. Policymakers tend to be quite 'old school' and cautious, so here we probably also care about the type I error more than type II error.
- * A test to conduct is to see if the bags of chocolate weigh 500 grams. We can look at the mean weight and see if it's significantly different from 500. Here, the manufacturer should care about type II error more. This is because focusing on type II error will allow them to identify flaws in their quality control process and ensure that they're not falsely marketing their product. From outsider's point of view (i.e., consumers), it is better if the company puts in place more stringent methods to ensure that they're giving us enough of the product. The company also has a lot to lose (their brand and reputation which took years to build) if it is found that they're cheating their customers

3. **p-value**

(a) Definition: Given a realization of the test statistic $t = T(x_1, \dots, x_n)$, the *p-value* associated with t is:

- $\mathbb{P}(T(x_1, \dots, x_n) \geq t \mid H_0 \text{ is true})$ if $H_A : \mu > \mu_0$

- $\mathbb{P}(T(x_1, \dots, x_n) \leq t \mid H_0 \text{ is true})$ if $H_A : \mu < \mu_0$
 - $\mathbb{P}(|T(x_1, \dots, x_n)| \geq |t| \mid H_0 \text{ is true})$ if $H_A : \mu \neq \mu_0$
- (b) intuitively, it measures the probability of observing data as extreme or more extreme than the one you've collected, assuming the null hypothesis holds.
- (c) we reject the null hypothesis if the p-value is less than α , a given user-specified significance level (typically $\alpha = 0.05$)

4. Type-I error and Type-II error

- (a) $\mathbb{P}(\text{Type-I error}) := \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ holds})$; False Positive Rates; A scalar
- (b) $\mathbb{P}(\text{Type-II error}) := \mathbb{P}(\text{not reject } H_0 \mid H_0 \text{ does not hold, and the true parameter is } \mu') =: \beta(\mu')$; False Negative Rates; Is a function of on True parameter μ' .
- (c) Our procedure controls the $\mathbb{P}(\text{Type-I error})$ (being less than or equal to α) and the $\mathbb{P}(\text{Type-II error})$ decreases as n increases.
- Repeat the experiments T times, in which the null hypothesis holds, we will falsely reject H_0 among approximately $T \times \mathbb{P}(\text{Type-I error})$ experiments.
 - Repeat the experiments T times, in which the null hypothesis doesn't hold and the true parameter is μ' , we will falsely not reject H_0 among approximately $T \times \beta(\mu')$ experiments.
- (d) Trade-off between $\mathbb{P}(\text{Type-I error})$ and $\mathbb{P}(\text{Type-II error})$
- Smaller $\alpha \implies$ large $\mathbb{P}(\text{Type-II error})$.
 - Extreme case $\alpha = 0$: never reject the null hypothesis, $\mathbb{P}(\text{Type-II error})$ error is 1 under regular conditions.

5. **Example** A food manufacturing facility employs an automated canning apparatus for the purpose of packaging canned food items. The predetermined standard weight for each can is 500g. A sample comprising ten randomly selected cans underwent weight measurement, yielding the following observations:

495, 510, 505, 498, 503, 492, 502, 512, 497, 506.

Assume the weight of each can, denoted by X is normally distributed $X \sim N(\mu, \sigma^2)$ with $\sigma = 6.5$. Is the machine working normally? (take significant level to be $\alpha = 0.05$)

- **Solution:** The test problem is

$$H_0 : \mu = 500 \leftrightarrow H_1 : \mu \neq 500.$$

The test statistics is

$$T(X_1, \dots, X_n) = \frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma},$$

where $n = 10$, $\sigma = 6.5$, $\mu_0 = 500$, $z_{0.025} = 1.96$, and $\bar{x} = 502$. The rejection region is

$$R = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, +\infty).$$

Thus,

$$\left| \frac{\sqrt{n}(\bar{x} - \mu_0)}{\sigma} \right| = 0.973 < 1.96,$$

meaning that the test statistics does not belong to the rejection region, and therefore, with $\alpha = 0.05$, we accept H_0 .